Mouse Dynamics for Hand Prediction Using Behavioral Biometrics

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Abstract
This study investigates whether mouse movement patterns can be used to predict which hand a user is using. A web-based drag-and-drop task was made to collect mouse dynamics data. Features such as trajectory curvature, movement efficiency, and direction changes were extracted and used to train machine learning models including Random Forest, Support Vector Machine (SVM), and XGBoost. Initial results revealed bias caused by time-based features, which were removed to improve model validity. The refined model achieved approximately 70% accuracy, demonstrating that movement-based features are more reliable for predicting hand usage.

1. Introduction
   Behavioral biometrics focuses on identifying users based on their behavior rather than physical characteristics. Mouse dynamics is a widely studied behavioral biometric that captures how users interact with a system through cursor movement, clicks, and drag actions.

Mouse movements contain patterns such as speed, trajectory, and curvature. These patterns can vary depending on how a user controls the mouse. This project explores whether these differences can be used to determine if a user is using their left or right hand.

The objective of this study is to design a system that collects mouse movement data and applies machine learning techniques to classify hand usage.

2. Research Questions

- Can mouse movement patterns be used to predict which hand a user is using?
- Do left and right hands produce different mouse behaviors?
- Can these differences be detected using machine learning?

3. Hypothesis
   Left and right hands produce different movement patterns, and these patterns can be used by machine learning models to classify hand usage.

These differences include:

- trajectory curvature
- movement efficiency
- direction behavior

4. Literature Review / Related Work
   Mouse dynamics has been widely studied as a behavioral biometric for authentication and user identification. Early work by Ahmed and Traore (2007) demonstrated that mouse movement patterns can uniquely identify users using neural network models. Their study achieved low false acceptance and rejection rates, showing that mouse behavior can act as a continuous biometric rather than a one-time login feature.

Later research by Zheng et al. (2011) showed that carefully engineered features combined with Support Vector Machines could achieve strong performance, including low error rates with relatively small amounts of interaction data. This suggests that meaningful behavioral patterns can be extracted even from short user sessions.

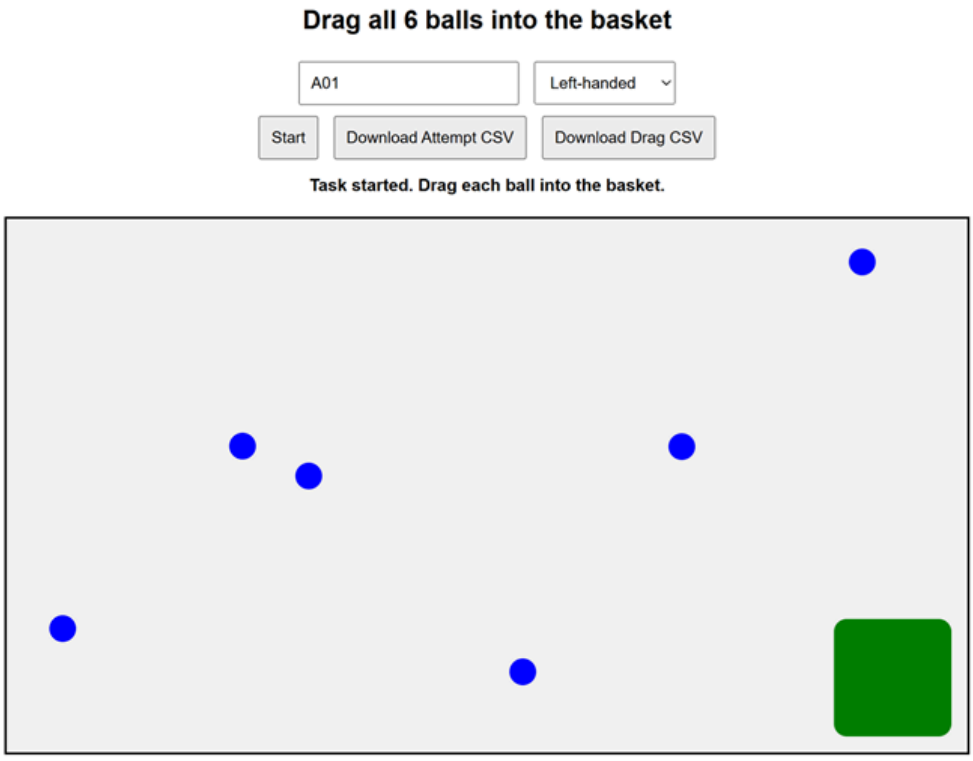
More recent studies using the Balabit Mouse Dynamics dataset have focused on action-based segmentation. Antal and Egyed-Zsigmond (2019) showed that separating mouse actions into movement, point-and-click, and drag-and-drop improves classification performance. Their work also identified drag-and-drop actions as highly discriminative, which directly supports the design of this project.

Additional studies by Almalki et al. (2021, 2023) reported high accuracy using machine learning models such as K-Nearest Neighbors and Random Forest. However, they also noted that results can vary depending on dataset size and evaluation setup.

Finally, survey research by Khan et al. (2024) highlights that mouse dynamics remains a promising but challenging biometric. Performance depends heavily on feature selection, data quality, and experimental design. This project builds on these findings by focusing on reducing bias and emphasizing movement-based features rather than time-based metrics.

5. Dataset

5.1 Data Collection
Data was collected using a custom web-based application where users were asked to drag six balls into a basket. This task captures natural mouse movement behavior.



5.2

Dataset Structure
Two datasets were initially collected:

- Attempt-level dataset (one row per session)
- Drag-level dataset (one row per drag action)

The drag-level dataset was later removed to simplify analysis and focus on overall behavior.

5.3 Features Collected
A total of 26 features were collected and grouped into:

- Task Performance
- Distance & Efficiency
- Speed & Acceleration
- Smoothness
- Movement Shape
- Control & Corrections
- Drag Behavior

6. Data Issues and Bias

6.1 Bias in the Data

Most participants were right-handed. So when these users used their left hand, their movements were slower and less accurate. As a result, the model incorrectly learned that slower movement indicates left-hand usage.

6.2 Fixing the Bias

To address this issue:

- time-based features were removed
- the model focused on movement shape features

This improved the validity of the model.

6.3 Data Collection Error

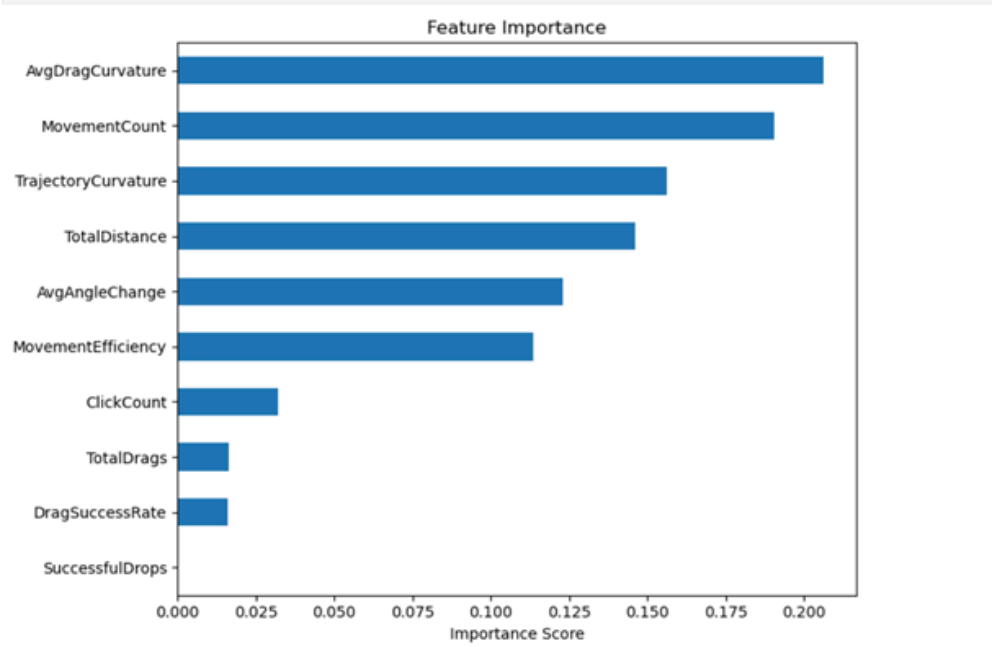
Enhanced pointer precision was not disabled in windows during data collection, this introduced inconsistencies in movement behavior. As a result the data collected up to that point was deleted completely.

7. Methodology

7.1 Feature Selection

The most important features used in the final model were:

- AvgDragCurvature
- MovementCount
- TrajectoryCurvature
- TotalDistance
- AvgAngleChange
- MovementEfficiency
- ClickCount
- TotalDrags
- DragSuccessRate
- SuccessfulDrops



These features were selected because they represent movement shape and control rather than speed, reducing bias and improving the interpretability of the model.

7.2 Models Used
The following models were implemented:

- Random Forest
- Support Vector Machine (SVM)
- XGBoost

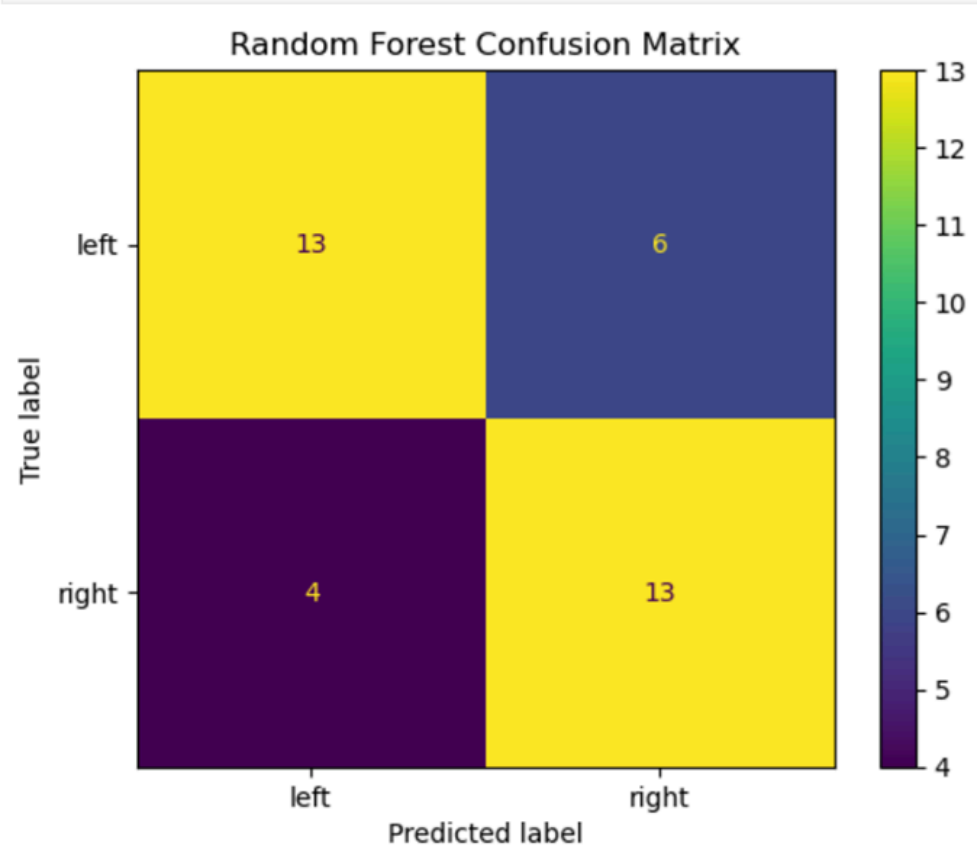
These models were selected because they can handle non-linear relationships in behavioral data but they differ in how they learn patterns from the data. Random Forest is an ensemble model that builds multiple decision trees and combines their results, making it effective for capturing complex, non-linear relationships in behavioral features. Support Vector Machine (SVM) works by finding an optimal boundary between classes, which can be effective when the data is well-separated but may struggle when patterns are highly irregular. XGBoost is a boosting-based model that builds trees sequentially to correct previous errors, making it powerful but more sensitive to noise and smaller datasets.

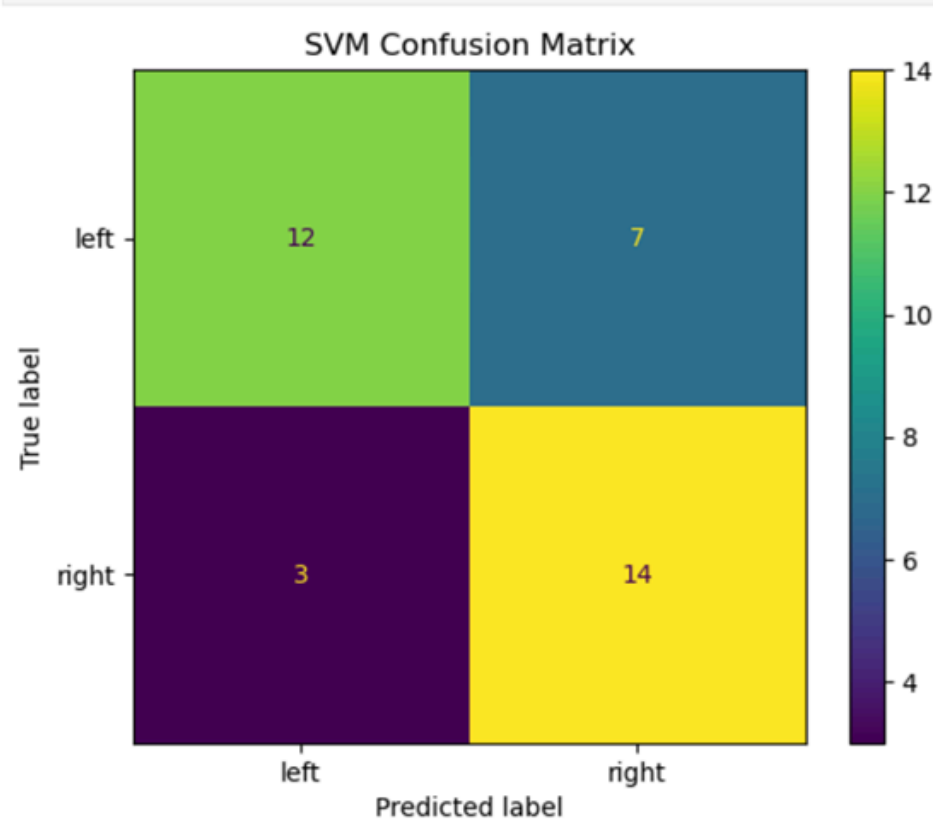
In this project, Random Forest and SVM performed similarly, while XGBoost showed slightly lower accuracy. This may be because the dataset is relatively small and contains noise, which can reduce the effectiveness of more complex models like XGBoost.

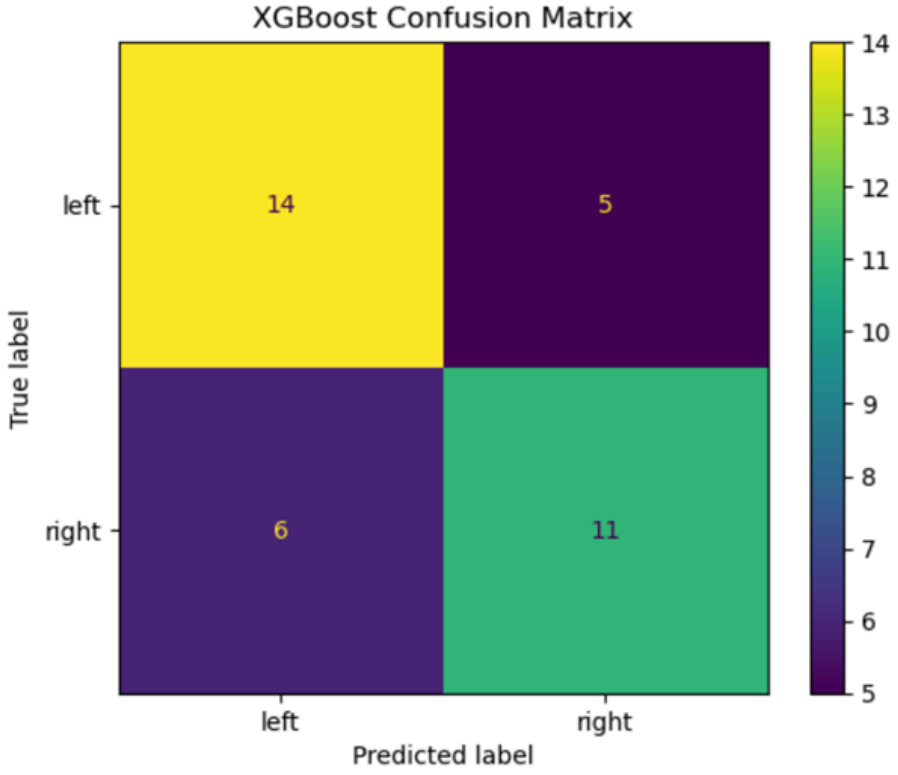
7.3 Model Process
Our workflow included:

- data cleaning
- feature selection
- model training
- evaluation using test data

8. Results
   The models achieved moderate accuracy of approximately 70%. Although the accuracy is moderate, it reflects the difficulty of the problem. Mouse dynamics is a behavioral biometric, which means the data is naturally noisy and varies between users. Unlike traditional datasets, there is no clear separation between classes, making perfect classification unlikely.




SVM Confusion Matrix
left
right
True label
12
7
3
14
left
right
Predicted label
14
12
10
8
6
4


XGBoost Confusion Matrix
left
right
True label
14
5
6
11
left
right
Predicted label
14
13
12
11
10
9
8
7
6
5

	Model	Test Accuracy
0	Random Forest	0.722222
1	SVM	0.722222
2	XGBoost	0.694444

9. Analysis of Results

The results show that movement shape features are more effective than time-based features for predicting hand usage. Features such as trajectory curvature and angle change capture how a user naturally controls the mouse, making them more reliable indicators than speed.

The confusion matrices reveal that the model makes more errors when predicting left-hand usage compared to right-hand usage. This is likely due to dataset imbalance, as most participants were right-handed. As a result, the model is slightly biased toward predicting right-hand usage.

Additionally, Random Forest and SVM achieved similar performance, while XGBoost performed slightly worse. This may be because XGBoost is more sensitive to noise and requires larger datasets to perform optimally. In contrast, Random Forest is more robust to noise and works well with smaller datasets, which explains its strong performance in this project.

Another key observation is that removing time-based features reduced overall accuracy but improved model reliability. This shows that higher accuracy does not always mean better performance. A model that relies on meaningful, unbiased features provides more valid and interpretable results.

Overall, the results suggest that while mouse dynamics can be used to predict hand usage, the problem remains challenging due to variability in human behavior. Additional data and improved feature engineering would likely improve performance.

10. Prediction System
    A web-based application was developed to:

- collect mouse movement data
- extract features in real time
- predict hand usage

The system uses a drag-and-drop interface and processes user input to generate predictions.

11. Conclusion
    This study demonstrates that mouse dynamics can be used to predict hand usage. While accuracy is moderate, the results support the hypothesis that movement patterns differ between hands. Movement-based features provide more reliable insights than speed-based features.Overall, the results were not as high as expected, but they still show that the approach is promising.

12. Future Work

- collect more data
- include more left-handed users
- improve feature engineering
- test additional models
- use controlled environments

13. References

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